MILITARY SPECIFICATION

AMPLIFIERS, SYNCHRO SERVICE), 6 WATT 400 CYCLE

This specificaticm has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification describes the requirements for the procurement of two types of 6 Watt 400 Cycle Synchro Servo Amplifiers, BUWEPS LD-499702 and BUWEPS LD-494773. They are designed to drive a size-11 or -15400 cycle servo motor. The amplifiers are identical except for sealing compound. These amplifiers consist of two transistorized push-pull stages driven by the output of an incorporated transistorized preamplifier.

1.2 Classification. - The amplifiers shall be of the following types:

Type I	**BUWEPS LD-499702,**	**400 Cy 6W Synchro Servo Amplifier Assembly**
Type II	**BUWEPS LD-499773,**	**400 Cy, 6W Synchro Servo Ampl, Assembly, (Sealed)**

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, or such portion thereof as may be designated herein, of the issue in effect on the date of invitation for bids, form a part of this specification.

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SPECIFICATIONS

FEDERAL

UU-P-268	Paper, Kraft, Untreated, Wrapping
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-901	Shockproof Equipment Class HI (High Impact) Shipboard Application, Tests for
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-L-10547	Liner, Case, Waterproof
MIL-P-17555	Preparation for Delivery of Electronic, Electrical and Electro-Mechanical Equipment and Associated Repair Parts
MIL-F-18870	Fire Control, Optical and Associated Equipment Intended for Shipboard Use, the General Specification for Design and Procurement of
MIL-W-21927	Weapons, Handling and Preparation for Delivery of; General Requirements for

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STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167 .	Mechanical Vibration of Shipboard Equipment

DRAWINGS

NAVY DEPARTMENT

BUWEPS LD-499702	400 Cy 6 W Synchro Servo Amplifier Assembly
BUWEPS LD-499773	400 Cy, 6W Synchro Servo Ampl, Assembly, (Sealed)
BUWEPS SA-2530405	Amplifier Test Fixture

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

NAVWEPS OS 6341	Miscellaneous General Ordnance Design Requirements
NAVWEPS OD 12352	Description and Operating Instructions for Amplifier Test Fixture Shop Accessory SA 2530405

(Copies of specifications, standards, drawings, and publications requested by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.) .

2.2 Other Publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation to bid shall apply.

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NATIONAL BUREAU OF STANDARDS PUBLICATION

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Ill.)

AMERICAN TRUCKING ASSOCIATION, INC. , PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th Street, N. W. , Washington 6, D. C.)

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Pre-production Samples. - Unless otherwise specified in the contract or order, pre-production samples of the amplifier shall be manufactured using the methods and procedures proposed for the production lot. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's methods are capable of yielding items that comply with the technical requirements of the contract.

3.1.2 Materials. - Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the amplifier shall conform to the detailed requirements given on the drawings and specifications listed in BUWEPS LD-499702 or BUWEPS LD-499773, and in Specification MIL-F-18870.

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3. 1,3 Finishes, Coatings, Construction, and Dimensions. - Finishes, coatings, construction, and dimensions for the amplifier shall be in accordance with the applicable drawings, specifications, and publications listed in BUWEPS LD-499702 or BUWEPS LD-499773.

3,1.4 **Moisture, Fungus, and Corrosion. - The amplifier shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on drawings and specifications in BUWEPS LD-499702 or BUWEPS LD-499773.**

3.1.5 **Marking. - Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in BUWEPS IJD-499702 or BUWEPS LD-499773,**

3.1.6 **Interchangeability. - Component parts of the assembled units (unless otherwise specified on the applicable drawings) fabricated in accordance with their applicable drawings shall be interchangeable and shall assemble with mating components of the units without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required, then changes in dimensions are necessary, and revision and correction of the applicable drawings is required. See 3.1.10.**

3.1.7 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the materials to which this specification applies will meet the requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor*s inspection gages. However, such procedure shall not relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.1.8 **Threads, - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings, and the NOT- GO gaging practice shall be as set forth in the aforementioned Handbook H28.**

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3.1.9 Requests for Deviation. - Each request that a deviation be authorized shall have been submitted to the Procuring Activity, and the deviation shall have been authorized by the Bureau of Naval Weapons or by a Naval Activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The deviation shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the Procuring Activity to evaluate its effect on performance and reliability, and the reasons for the suggested deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the departure, the identifying number of the item, the contractor (or bidder) and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items, or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.

3.1.10 Conflicting Requirements. - Conflicting requirements arising between this specification or any specification, publication, or drawing listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.2 Definitions. - Definitions applicable to the environmental requirements of this specification are as follows:

3.2.1 Tropical Environment. - Tropical environment is defined as conditions of high humidity, high temperature, and spore exposure, which present an opportunity for fungi growth.

3.3 Performance Requirements and Product Characteristics. - The 6 watt 400 cycle synchro servo amplifier shall meet the following performance requirements and product characteristics:

3.3.1 Voltage Gain. - When tested with the special test equipment specified in 4.3 the voltage gain of the amplifier shall be such as to produce a vacuum tube voltmeter indication at the amplifier output of 22.0 plus or minus 0.5 volt alternating current when the input signal to the amplifier is between 0.20 and 1.00 volt alternating current.

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3.3.2 Noise. - When tested with the special test equipment specified in 4.3 the noise level of the amplifier with the input shorted shall be such as to produce a vacuum tube voltmeter indication of less than 2.50 volts alternating current.

3.3.3 Saturation. - When tested with the special test equipment specified in 4.3 the output of the amplifier shall be between 31.0 and 37.0 volts alternating current when the input signal indication is 60.0 plus or minus 0.5 volt alternating current.

3. 3.4 Power drain.

3.3.4.1 Input Signal at 1.0 Volt, Rotor Shaft Locked. - When tested with the special test equipment specified in 4.3 the power drain of the amplifier shall be such as not to produce a meter indication greater than the following when the input signal to the amplifier is 1.000 volts plus or minus 0.025 volt alternating current.

Circuit Identification	Power Drain Indication
1 volt direct current	5.0 milliamperes (maximum)
28 volts (peak)	180.0 milliamperes (maximum)
30 volts direct current	22.0 milliamperes (maximum)
48 volts (peak)	9.0 milliamperes (maximum)

3.3.4.2 Input Signal Shorted, Rotor Shaft Locked. - When tested with the special test equipment specified in 4.3, the power drain of the amplifier shall be such as not to produce a meter indication greater than the following when the input signal to the amplifier is shorted.

Circuit Identification	Power Drain Indication
1 volt direct current	4.0 milliamperes (maximum)
28 volts (peak)	50.0 milliamperes (maximum)
30 volts direct current	22.0 milliamperes (maximum)
48 volts (peak)	12.0 milliamperes (maximum)

3.3.5 Coarse-to-fine switching.

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3.3.5.1 36 Speed Switch - Coarse Synchro Input Signal 0.5 Volt Level. - When tested with the special test equipment specified in 4.3, the 36 speed coarse to fine switching action of the amplifier shall be such that when the input signal is 0.500 plus or minus 0.025 volt, the output signal shall not exceed 3.00 volts.

3.3.5.2 36 Speed Switch - Coarse Synchro Input Signal 2,0 Volt Level. - When tested with the special test equipment specified in 4.3, the 36 speed coarse to fine switching action of the amplifier shall be such that when the input signal is 2.000 plus or minus O.025 volt, the output signal shall be 20.0 volts minimum.

3.3.5.3 10 Speed Switch - Coarse Synchro, Input Signal 6.0 Volt Level. - When tested with the special test equipment specified in 4.3, the 10 speed coarse to fine switching action of the amplifier shall be such that when the input signal is 6.00 plus or minus O. 50 volt, the output signal shall not exceed 3.00 volts.

3.3.5.4 10 Speed Switch - Coarse Synchro, Input Signal 9.0 Volt Level. - When tested with the special test equipment specified in 4.3, the 10 speed coarse to fine switching action of the amplifier shall be such that when the input signal is 9.00 plus or minus 0.05 volt, the output signal shall be 20.0 volts minimum,

3.4 Environmental Requirements. - The 6 watt 400 cycle synchro servo amplifier shall meet the other applicable requirements of this specification during or after tests defined or specified in 4.5.

3.4.1 Vibration, Operating Condition. - The amplifier shall withstand the Type I vibration requirements specified in Standard MIL-STD-167, shall function during and after vibration tests, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.2 Vibration-Transportation. - The amplifier after being subjected to the tests specified in 4.5.2 shall show no damage and shall meet all the requirements of Section 3.

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3.4.3 Shock, Non-operating Condition. - The amplifier shall withstand the Type A shock requirements specified in Specification MIL-S-901, and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.4 Shock Transportation, Storage and Handling. - The amplifier after being subjected to the tests specified in 4.5.4 shall show no damage and shall meet all the requirements of Section 3.

3.4.5 Temperature, Operating Condition. - The amplifier shall withstand operational ambient temperatures of between zero degrees Fahrenheit (minus 18 degrees centigrade) and 185 degrees Fahrenheit (85 degrees centigrade) in accordance with Specification MIL-F-18870, shall function during and after temperature tests, and thereafter shall show no damage or corrosion and shall meet all the performance requirements and product characteristics of this specification.

3.4.6 Temperature, Non-operating Condition. - The amplifier shall withstand storage temperatures of between minus 40 degrees Fahrenheit (minus 400 degrees centigrade) and 160 degrees Fahrenheit (7 I degrees centigrade) in accordance with Condition II of NAVWEPS OS 6341, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification. .

3.4.7 Humidity. - The amplifier shall withstand operational ambient humidities in accordance with Specification MIL-F-18870, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.8 Tropical Environment. - The amplifier shall withstand the tropical environmental conditions specified in Specification MIL-F-5272 under "Fungus Resistance Tests, ” and thereafter shall show no fungus growth, corrosion, or other damage, and shall meet all the performance requirements and product characteristics of this specification.

3.5 Clean-Up. - Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

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3.6 Workmanship. - The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to "neatness and thoroughness of soldering, marking of parts and assemblies, riveting, machine screw assemblage, and freedom of parts from burrs and sharp edges.

3.6.1 The standards of workmanship exhibited in any approved pre-production sample, subject to any qualification stated in the Governments notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Pre-production Sample. - Unless otherwise specified for each contract or order, a representative sample of three amplifiers shall be manufactured by the methods and procedures proposed for the production lot. These shall be submitted for pre-production tests at an activity designated by the Bureau of Naval Weapons, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity of the completion of tests on the pre-production samples shall be at the contractor?s risk. Pre-production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

4.1.2 Periodic Production Sampling. - Periodic production samples shall be selected from units that have satisfactorily passed the acceptance inspections of 4.4. These samples shall be subjected to the tests of 4.5 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

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Step 1. Upon initiation of production, randomly select one (1) unit from each group of ten (10) units. Upon Satisfactory Compliance of samples selected from three (3) consecutive groups, shift to Step 2.

Step 2. Randomly select one (1) unit from each group of twenty-five (25) units. Upon satisfactory compliance of samples from three (3) consecutive groups, shift to Step 3.

Step 3. Randomly select one (1) unit from each group of fifty (50) units. Continue with Step 3 for the balance of the contractor until a defective unit is found.

When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

a. Determine the cause and effect.

b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

Return to Step 1 and proceed as indicated when inspection is resumed.

4.2 Classification of Tests. - The inspection and testing of the amplifier shall be classified as follows:

4.2.1 Acceptance Tests. - These tests shall be accomplished on each amplifier submitted for acceptance under contract. These tests are detailed in 4.4.

4.2.2 Pre-production and Periodic Production Tests. - Pre-production and periodic production tests shall be accomplished on samples selected as specified in 4.1.1 and 4,1.2, which are representative of the production of the amplifier after the award of the contract to determine that the production meets the requirements of this specification. These tests are detailed in 4.5. Acceptance shall be based on no defects in the sample.

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4.2.3 Tests After Delivery. - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all the tests specified herein, as deemed necessary, after installation of the amplifier. Any defects discovered during or after these tests due to the contractor?s failure to comply with drawings or specifications for this equipment shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.3 Test Equipment. - The following item of test equipment is required to perform the acceptance tests set forth in this specification:

4.3.1 Shop Accessory SA-2530405. Amplifier Test Fixture.

4.4 Acceptance tests.

4.4.1 The contractor shall conduct acceptance tests as specified in 4.2.1 to assure that the amplifier is in compliance with the requirements of 3.1, 3.3, 3.5, and 3.6. Manufacturer's certification of materials specified in the applicable drawings shall be used in lieu of material tests.

4.4.2 Test Conditions. - Unless otherwise specified, the amplifier shall be subjected to acceptance tests under the following conditions

4.4.2.1 Temperature: Room ambient 65 to 95 degrees Fahrenheit (18 to 35 degrees centigrade).

4.4.2.2 Altitude: Normal ground atmospheric pressure.

4.4.2.3 Vibration: None.

4.4.2.4 Humidity: Room ambient to 95 per cent relative maximum.

4.4.3 Test and Inspection Equipment and Facilities. - The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

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4.4.4 Test Procedure. - The amplifier shall be tested in accordance with the detailed test procedure given in NAVWEPS OD 12352 to determine compliance with the requirements of 3.1, 3.3, 3.5, and 3.6.

4.4.4.1 Preservation, Packing and Packaging. - Preservation, packing and packaging of the amplifier shall conform to Section 5 of this specification.

4.5 Pre-production and periodic production sample tests.

4.5.1 Vibration, Operating Condition. - The amplifier, rigidly mounted in a suitable vibration fixture, shall be subjected to vibration tests so as to meet the requirements specified in 3.4.1, and shall meet all the performance requirements and product characteristics of this specification during and after the vibration tests.

4.5.2 Vibration-Transportation. - The amplifier packaged and packed as specified in Section 5, shall be vibration tested as specified in Specification MIL-W-21927.

4.5.3 Shock, Non-operating Condition. - The amplifier, rigidly mounted in a suitable shock fixture, shall be subjected to shock tests so as to meet the requirements specified in 3.4.3, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.4 Shock-Transportation, Storage and Handling. - The amplifier packaged and packed as specified in Section 5, shall be subjected to the appropriate rough handling tests specified in Specification MIL-W-21927.

4.5.5 Temperature, Operating Condition. - The amplifier shall be subjected to temperature tests so as to meet the requirements specified in 3.4.5, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.6 Temperature, Non-operating Condition. - The amplifier shall be subjected to temperature tests so as to meet the requirements specified in 3.4.6, and thereafter shall meet all the performance requirements and product characteristics of this specification.

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4.5.7 Humidity. - The amplifier shall be subjected to humidity tests so as to meet the requirements specified in 3.4.7, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.8 Tropical Environment. - The amplifier shall be exposed to fungus tests so as to meet the requirements specified in 3.4.8, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.6 Failure of the amplifier to meet any of the requirements and tests of this specification shall be considered cause for rejection.

4.7 Functional Tests. - At the conclusion of the tests specified in 4.5.2 and 4.5.4 above, functional and other tests, as applicable, shall be conducted on the amplifier to determine freedom from operational malfunction and compliance with all requirements of Section 3.

4.8 Waiver of Tests. - Tests specified in 4.5.2 and 4.5.4 above may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. - In addition to the following, Level A packaging shall be in accordance with Specification MIL-P-17555.

5.1.1.1 Cleaning, Drying and Preservative Application. - Not applicable.

5.1.1.2 Unit Packaging. - Unit packaging of the amplifier shall be in accordance with Method IIe of Specification MIL-P-116. The unit package shall contain one (1) amplifier packaged in a fiber box in accordance with Specification PPP-B-636, Class 2, with Kraft paper overwrap in accordance with Specification UU-P-268.

5.1.1.3 Intermediate Package. - Not applicable.

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5.1.1.4 Cushioning. - The unit package and pack shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to ensure that the amplifier will meet all the requirements of Section 3 after being tested in accordance with 4.5.2 and 4.5.4.

At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 per cent. The pH range of the material shall be between 6.0 and 8.0.

5.1.2 Level B. - Not applicable.

5.1.3 Level C.

5.1.3.1 Cleaning, Drying and Preservative Application. - Not applicable.

5.1.3.2 Unit Packaging. - Unit packaging of the amplifier shall be in accordance with Method III of Specification MIL-P-116. The unit package shall contain one (1) amplifier.

5.1.3.3 Intermediate Package. - Not applicable.

5.2 Packing.

5.2.1 Level A. - In addition to the following, Level A packing shall be in accordance with Specification MIL-P-17555.

5.2.1.1 Exterior Containers; - Amplifiers packaged as described in 5. 1.1.2 shall be packed in a wood box in accordance with Specification PPP-B-601, overseas type, or PPP-B-621, Class 2 (gross weight not to exceed 150 lbs), or in fiber boxes in accordance with Specification PPP-B- 636, Class 2 (gross weight not to exceed 70 lbs).

5.2.1.2 Case Liners. - Wood exterior containers shall be provided with case liners in accordance with Specification MIL-L-10547.

5.2.2 Level B. - In addition to the following, Level B packing shall be in accordance with Specification MIL-P-17555.

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5.2.2.1 Exterior Containers. - Amplifiers packaged as described in 5,1.1.2 shall be packed in a wood box in accordance with Specification PPP-B-601, domestic type, or PPP-B-621, Class 1 (gross weight not to exceed 150 lbs), or in fiber boxes in accordance with Specification PPP-B-636, Class 1 (gross weight not to exceed 70 lbs).

5.2.2.2 Case Liners. - Wood exterior containers shall be provided with case liners in accordance with Specification MIL-L-10547.

5.2.3 Level C. - Amplifiers packaged in accordance with 5.1.3.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Special Markings. - None, unless otherwise specified.

5.3.2 Normal Markings. - In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended Use. - The 6 watt 400 cycle synchro servo amplifier is intended for use in the Angle Solvers Mark 18 Mod 2 and Mod 3, the Fire Control System Mark 112 Mod 1, the Sonar Receiving Sets AN/BQG-1, -2, -3, -4, or other suitable application.

6.2 Ordering Data. - Procurement documents should specify the specific title, number, and date of this specification and exceptions to this specification; applicable drawings and other documents; selections of applicable levels of packaging and packing; number of unit containers per pack; and guarantee clause. (See 6.5.)

6.3 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

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For Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or in domestic locations for an indefinite period.

For Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116 and to para. 4.8 of this specification.

6.5 Guarantee. - The contract or order shall include a requirement for a guarantee stating that any amplifier which fails to meet the requirements of this specification within one year after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

6.6 The contract or order shall include a sufficient number of extra items to cover production and pre -production testing samples.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-A-22334(Wep) AM PLIFIERS, SYNCHRO SERVO 6 WATT 400 CYCLE

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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DEPARTMENT OF THE NAVY
Bureau of Naval Weapons
Washington, D. C. 20360

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